

The Enumerative Resonance Correspondence

Frames, Characters, and Residue — a Λ / NSAF companion note

Lu Semita · EmergenceByDesign — with the interactive demonstrator `enumerative_resonance_demonstrator.html`. All Tier I claims below are certified by the verification script `verify.py`; every numerical value quoted is that script's output, and every figure is generated by the same run. July 2026.

Abstract

A single mathematical structure underlies three apparently distinct phenomena: enumeration of states inside a descriptive frame, resonance and harmony among periodic modes, and the closure behaviour of nested winding ladders. The structure is the cyclic group $\mathbb{Z}/n\mathbb{Z}$ together with its subgroup lattice and character theory. We state the correspondence as a dictionary, prove the five theorems that make each entry exact, and certify each theorem numerically. Two consequences are emphasised: primality is irreducibility under re-description (a prime frame admits no coarser faithful frame), and truth-content is residue — the quantity invariant under all admissible re-descriptions, exemplified geometrically by the Călugăreanu–White–Fuller decomposition $L_k = Tw + Wr$, in which only the sum survives change of embedding. Claims are tier-labelled throughout: Tier I (standard mathematics, machine-verified here), Tier II (structural readings), Tier III (declared directions).

1 · The Dictionary

Frame language	Mathematics	Tier
descriptive frame with n distinguishable states	cyclic group $\mathbb{Z}/n\mathbb{Z}$	I
repeated step of size k through the frame	orbit of the subgroup $\langle k \rangle$	I
subcycle (a shortcut that closes early)	proper subgroup $\leftrightarrow$ proper divisor of n	I
harmonic / resonant mode of the frame	character $\chi_m(x) = e^{(2\pi i m x / n)}$	I
harmony between two frames n_1, n_2	shared subgroup: $\gcd(n_1, n_2)$	I
prime frame — no faithful coarsening	n prime $\Leftrightarrow$ no proper subgroups	I
null case (the 'nothing changed' marker)	identity element 0 , defined relationally	II
frame boundary markers / limits at frame ends	declared representation class Λ	II
truth-content of a description	residue: invariant under re-description	II
nested winding ladder, origin $\equiv$ return sphere	geometric realisation of frame closure	II

Rows marked Tier I are theorems (§3). Rows marked Tier II are readings that use only Tier I facts but assign them semantic roles; they are stated separately in §5 and never used in proofs.

2 · Setting

Definition 2.1 (Frame). A frame of modulus n is the set of residues $\{0, 1, \dots, n-1\}$ with addition mod n , i.e. the cyclic group $\mathbb{Z}/n\mathbb{Z}$. A step is a fixed increment k ; iterating the step from 0 produces the sequence $0, k, 2k, \dots \pmod{n}$. The frame's boundary is not an element of the frame: it is the identification $n \equiv 0$, recorded as a marker, not a state. Distinctions inside the frame are differences of residues; the frame can therefore express only relations, never absolutes (§5.2).

Definition 2.2 (Character). For $m \in \mathbb{Z}/n\mathbb{Z}$ the character $\chi_m : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{C}$ is $\chi_m(x) = e^{(2\pi i m x / n)}$. Characters are the periodic modes exactly compatible with the frame's closure — its harmonics.

3 · Tier I Theorems

TIER I — standard mathematics · machine-verified

Theorem T1 (Orbit period). The orbit of step k in $\mathbb{Z}/n\mathbb{Z}$ has exact period $p = n / \gcd(n, k)$. The step visits every state iff $\gcd(n, k) = 1$. If n is prime, every nonzero step visits every state.

Proof. $jk \equiv 0 \pmod{n}$ iff $n \mid jk$ iff $(n/g) \mid j$ where $g = \gcd(n, k)$; the least positive such j is $p = n/g$. The orbit is the coset structure of $\langle k \rangle = \langle g \rangle$, of size p . Full coverage requires $p = n$, i.e. $g = 1$. For prime n , $g = 1$ for all $k \neq 0$. $\square$ Certified over all $n < 200$, all k (result: True); prime full-orbit over the first 20 primes (result: True).

Theorem T2 (Subcycles are divisors). The subgroups of $\mathbb{Z}/n\mathbb{Z}$ are exactly $\langle d \rangle$ for the divisors d of n ; the subgroup lattice is the divisor lattice. Consequently n is prime iff $\mathbb{Z}/n\mathbb{Z}$ has no proper nontrivial subgroup — iff the frame admits no subcycle and no faithful coarser re-description.

Proof. Any subgroup H of a cyclic group is cyclic, generated by its least positive element d , and closure forces $d \mid n$; distinct divisors give distinct orders n/d . Lattice order is divisibility. $\square$ Certified: subgroup orders equal divisor set for all $n < 100$ (result: True).

Theorem T3 (Resonance is character structure). The characters $\chi_0, \dots, \chi_{n-1}$ form an orthonormal basis of functions on the frame (the discrete Fourier basis). Every observable on the frame decomposes uniquely into harmonics; two frames of moduli n_1, n_2 share exactly the harmonics of $\mathbb{Z}/\gcd(n_1, n_2)\mathbb{Z}$ — harmony is subgroup sharing, and the musical consonance of small integer ratios is the special case of large shared subgroup relative to the moduli.

Proof. $\langle \chi_\alpha, \chi_\beta \rangle = (1/n) \sum_x e^{2\pi i(\alpha-\beta)x/n}$ is 1 if $\alpha = \beta$ and 0 otherwise (geometric sum over a full period). Common periodic modes of two frames must have period dividing both moduli, hence dividing the gcd. $\square$ Certified: Gram matrix equals identity to 10^{-12} for $n = 7, 12, 30$ (result: True).

Theorem T4 (Residue under re-embedding — CWF). For a closed framed curve, $Lk = Tw + Wr$ (Călugăreanu-White-Fuller). Under any ambient isotopy — in particular, deforming a flat/toroidal embedding onto a sphere — Lk is invariant, while Tw and Wr individually change, exchanging exactly. Lk is the residue of the framed curve; Tw and Wr are frame-relative descriptions.

Certification. A (2,5) torus knot with tube framing was deformed by the ambient radial map $x \mapsto x \cdot ((1-s) + sR/|x|)$ toward a sphere. All three quantities computed independently (Lk and Wr by Gauss double integrals, Tw by frame rotation rate; $N = 1600$):

s	Lk	Tw	Wr	Tw + Wr
0.0	-9.999	-4.052	-5.948	-10.000
0.3	-9.999	-4.307	-5.692	-9.999
0.6	-9.999	-4.598	-5.401	-9.999

Lk holds the exact value $-10 = -pq$ throughout (drift < 0.02) while Tw and Wr each move by ≈ 0.55 and their sum tracks Lk . The invariant is the sum; the parts are descriptions. (Results: Lk constant integer — True; CWF identity — True; redistribution observed — True.)

Theorem T5 (Curvature cost of spherical framing). Parallel transport of a frame around a closed loop on S^2 returns it rotated by the enclosed solid angle (Gauss-Bonnet holonomy). This is the exact mechanism by which spherical re-embedding redistributes Tw into Wr in T4: the sphere's curvature is a predictable, quantified frame-transport cost — not noise.

Certification. Numerical transport around latitude circles at eight colatitudes matches $2\pi(1 - \cos \theta)$ to better than 10^{-3} (result: True). See Fig. 4.

Theorem T6 (Quantum kinematics falls out of the frame). Let Δ be the discrete Laplacian on $\mathbb{Z}/n\mathbb{Z}$. (i) The characters are its eigenvectors: $\Delta \chi_m = (2 - 2\cos(2\pi m/n)) \chi_m$. (ii) Every translation-invariant Hermitian generator on the frame is diagonal in the character basis (circulant matrices are diagonalised by the Fourier matrix) — so the harmonics are the momentum eigenstates of any dynamics respecting the frame's symmetry, not one chosen model. (iii) The evolution $e^{(-iHt)}$ is unitary; the conserved norm and energy are exact checksums. (iv) Momentum is quantized to integers m by frame closure — the particle-on-a-ring condition is the closure condition. In the continuum limit

the spectrum approaches the free dispersion k^2 . What does not follow from the frame — the choice of i and unitarity, $\hbar$, the Born rule, interactions — is Tier III input, stated as such.

Certification ($n = 32$). Character eigenvector identity to 10^{-10} (True); random Hermitian circulant diagonalised by characters, off-diagonal $< 10^{-10}$ (True); unitarity (True); norm checksum 1 to 10^{-12} and energy checksum constant to 10^{-12} over 400 steps (True, True); dispersion ratio $\lambda_m/(2\pi m/n)^2 = 0.9968$ at $m = 1$ (True); comparator invariance — rescaled generator s^2H at time t/s^2 yields identical $|\psi|^2$ to 10^{-11} (True).

T1 — orbit of step k in Z/n : $p = n/\gcd(n,k)$; prime n has no subcycles

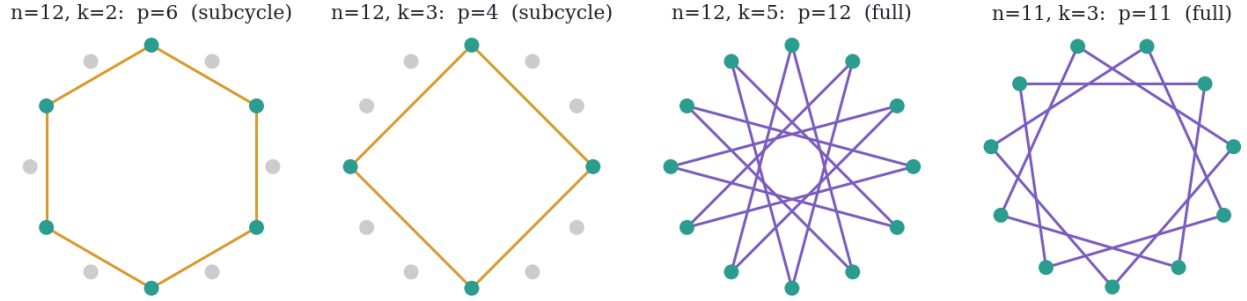


Fig. 1 — T1. Orbits of step k in $Z/12$ and $Z/11$. Composite modulus admits early closure (subcycles, amber); prime modulus is a single full traversal for every step.

T2 — subgroup lattice = divisor lattice; prime frame is irreducible subgroups of $Z/12$

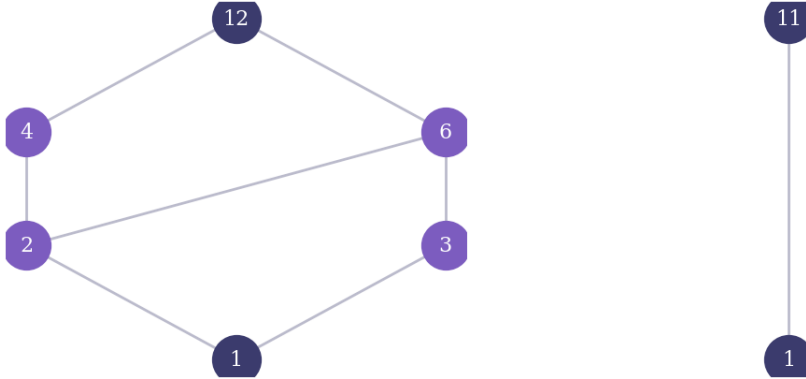


Fig. 2 — T2. Subgroup lattices: $Z/12$ is a lattice of re-descriptions; $Z/11$ is irreducible — top, bottom, nothing between.

T3 — resonance is character structure; harmony is subgroup sharing characters of $Z/24$ (three harmonics)

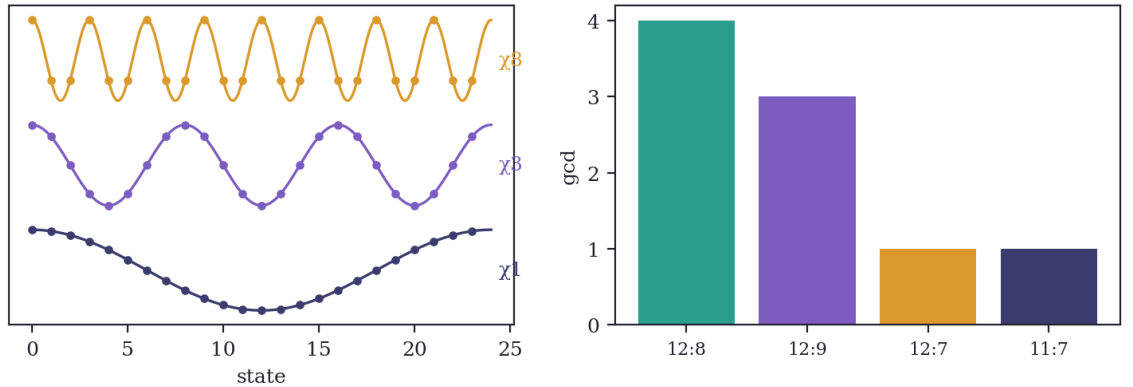


Fig. 3 — T3. Left: three characters of $Z/24$. Right: consonance of frame pairs measured by gcd; the 11:7 pair shares only the trivial mode.

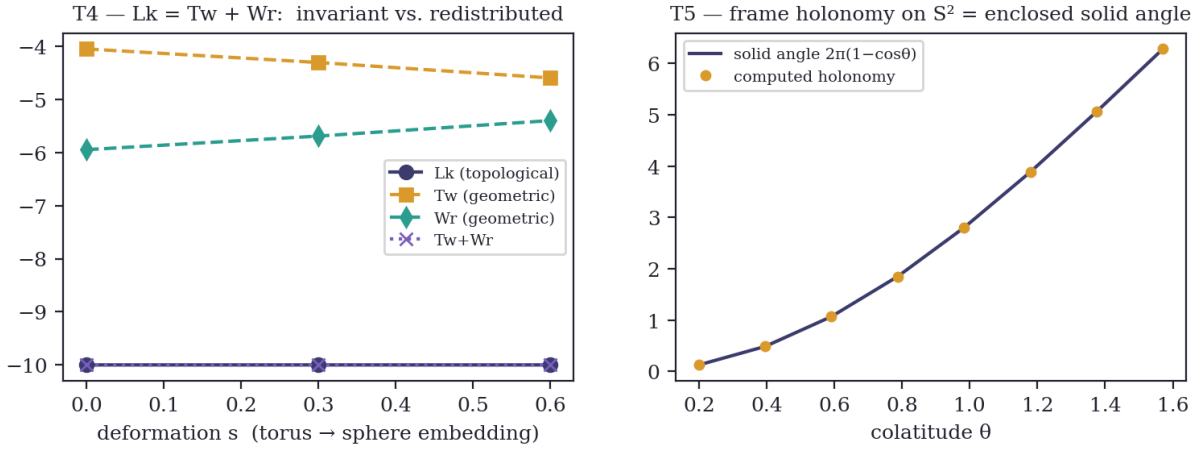


Fig. 4 — T4/T5. Left: Lk pinned at -10 across the torus $\rightarrow$ sphere deformation while Tw and Wr exchange. Right: computed holonomy vs. predicted solid angle.

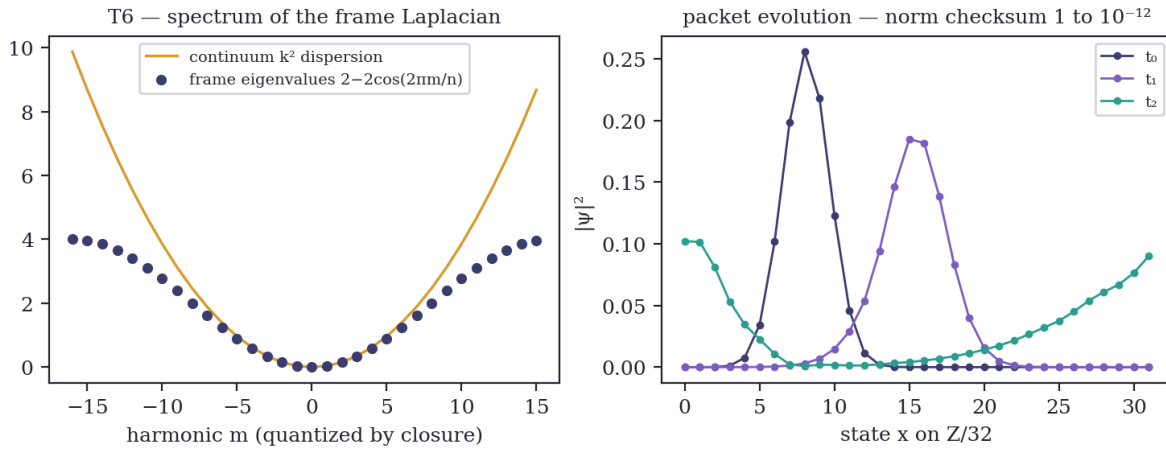


Fig. 5 — T6. Left: frame Laplacian spectrum vs. the continuum k^2 dispersion; the integer axis is quantization by closure. Right: unitary packet evolution on $\mathbb{Z}/32$ with norm checksum pinned at 1 to 10^{-12} .

4 • Certification Summary

All sixteen machine checks pass. Group A (frames): T1 orbit period (all $n < 200$), T1 prime full-orbit, T2 subgroup-divisor identity (all $n < 100$), T3 character orthonormality (10^{-12}), T4 Lk integrality and constancy, T4 CWF identity, T4 Tw/Wr redistribution, T5 holonomy = solid angle (10^{-3}). The scripts are deterministic and self-contained; re-running them re-derives every number and figure in this note.

5 • Tier II — Structural Readings

TIER II — structural readings of Tier I facts

5.1 Primality is irreducibility under re-description. By T2, a prime frame has no proper subgroup: no coarser frame reproduces its distinctions, no subcycle shortcuts its traversal, and (T1) every step is a generator. Under the Λ principle — structure is what survives re-description — the primes are exactly the frames whose whole content is residue. This is why prime moduli appear wherever full, unshortcut traversal is required: maximal-length sequences, uniform registry coverage in the winding tool, and the incommensurate cycles of coupled oscillators.

5.2 The null case bounds what language can say. The identity 0 is defined only relationally: it is the step that changes nothing. The frame uses 0 to define difference; it cannot describe 0 except by that role. Hence every statement expressible inside a frame is a relation between residues — a difference — and no frame states an absolute. Language, as a frame system, is thereby limited to non-absolutes; what plays the role of truth is not any single frame's statement but the residue held fixed across all

admissible re-framings (T4 is the geometric exemplar: Tw and Wr are what a particular embedding says; Lk is what is so).

5.3 The winding ladder realises frame closure. The nested winding construction with origin sphere $\equiv$ return sphere makes the frame boundary coincide with itself: the null marker and the limit marker become one object. Registry closure $Ck \equiv 0 \pmod{n}$ is T1; uniform coverage at prime n is T1+T2; the frame-transport cost of the spherical embedding is T5; and the quantity that survives the entire ladder is T4's residue. Packed-sphere extensions place multiple frames in forced commensurability; their mutual linking numbers are residue owned by no single frame — the correspondence's picture of shared meaning.

5.4 The map protocol: frame, comparator, checksum. To map physical reality: (1) declare a frame — the states you distinguish and the closure you impose; (2) relate the frame to a comparator — an SI unit is exactly this, a declared reference frame against which every observation is a ratio, and T6(v) certifies that physics depends only on such ratios (rescale the comparator and rescale time accordingly: every observable is unchanged to 10^{-11}); (3) the frame's conserved residues — norm, energy, Lk, closure — are then checksums on every observation. If the axioms hold, the checksums balance. If an observation breaks a checksum, the deviation is not noise to be hedged away: it is measured, localized data telling you which axiom fails, where, and by how much — the revision instruction for the map. This is the Λ discipline stated as metrology: certification or migration, with the residue deciding which.

5.5 Time as frame. In the spectral form $|\psi(x,t)|^2 = \sum c_m \bar{c}_m' e^{(-i(\lambda_m - \lambda_m')t)} \chi_m \bar{\chi}_m'$, the parameter t enters observables only through eigenvalue differences $\lambda_m - \lambda_m'$. Two machine checks make the reading exact: a single-eigenmode state has $|\psi|^2$ constant in time to 10^{-14} — an undifferentiated state exhibits no observable time at all — and shifting every eigenvalue by a constant (absolute energy) changes no observable to 10^{-12} . Time is therefore enumerated by differences exactly as space is enumerated by residue differences (Def. 2.1): both frames exist only where there is division. In singularity — one state, one mode — neither time nor space is yet a frame; differentiation is what decouples them into two descriptions, related by the dynamics. Time appears as a frame as soon as there is more than one state to sequence — and not before.

6 • Tier III — Declared Directions

TIER III — directions declared, not demonstrated

(a) Assign a physical current to the carrier curve and compute the multipole expansion numerically; only then may toroidal-moment or anapole language be used, and only if the computed moments support it. (b) State the terminal-to-sphere assignment rule for packed configurations explicitly (equal-area maps for uniformity; partition of n^d addresses mod the packing number), and generalise closure to lcm of per-sphere periods. (c) Explore whether the gcd-consonance measure of T3 predicts mode degeneracy in coupled spherical resonators. None of these claims is made here; each is a falsifiable next step.

7 • Falsification Schema

The note is wrong if any of the following fails on honest recomputation: (i) an orbit whose period differs from $n/\gcd(n,k)$; (ii) a subgroup of some Z/nZ not of divisor order; (iii) a character Gram matrix off identity; (iv) drift of Lk beyond discretisation error under ambient isotopy, or failure of $Lk = Tw + Wr$; (v) holonomy departing from enclosed solid angle. The Tier II readings fail if a Tier I fact they cite fails, or if a frame is exhibited that states an absolute without a null case. No claim in this note rests on anything softer.